

KKBox’s Challenge

MUSIC RECOMENDATION

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PROBLEM:

The rise of streaming platforms and the development of personalized recommendation systems have changed the way people discovered new music.

However it is rather difficult to accurately capture individual taste variability and to make reliable predictions when user data is limited or entirely new.

In response to these issues, this work aims to design an AI model that predicts whether a listener will replay a song within a defined period of 1 month, using historical interaction patterns to enhance music personalization and recommendation accuracy.

OBJECTIVES:

GENERAL:

To predict human behavior related to <Music Tastes> to provide a solution for algorithmic recommendation by applying Systems Thinking and Systems Sciences Foundations Knowledge.

SPECIFIC:

1) To apply Systems Thinking to the problem, to understand its elements and interactions, by analysing and applying concepts relevant on the subject.

2) To develop and train an AI model to predict if the user will replay the song by using the system model and the given dataset.

3) To evaluate the metrics of the model to analize its efectiveness by applying AUC-ROC.

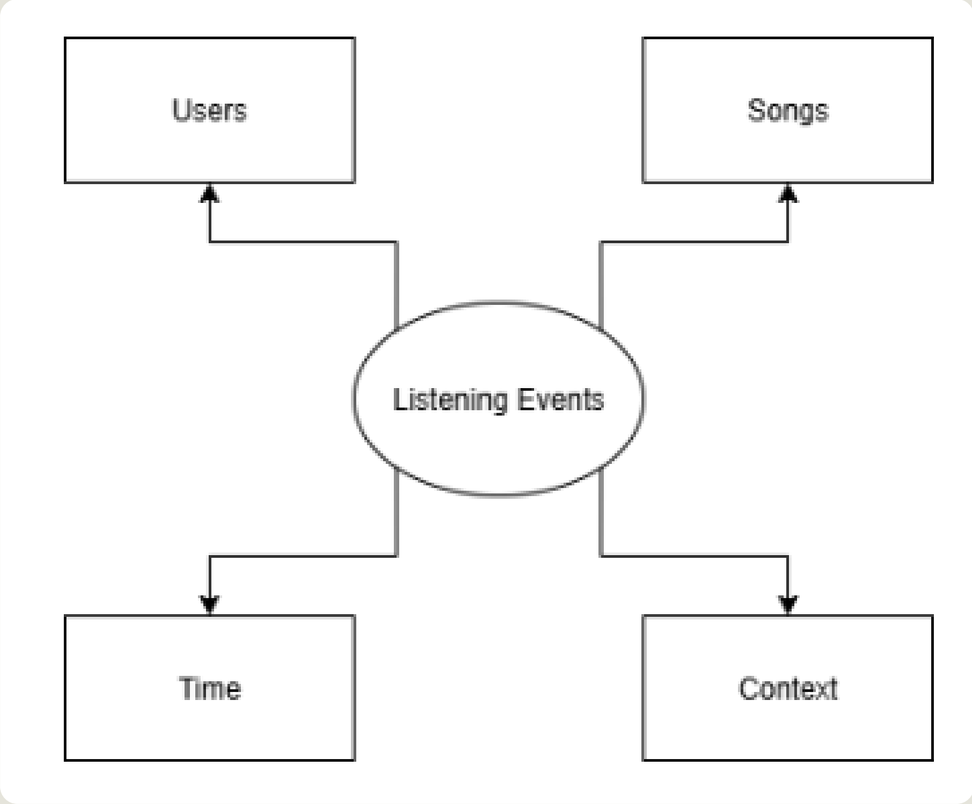
SYSTEM ANALYSIS:

Users: demographic and registration data.

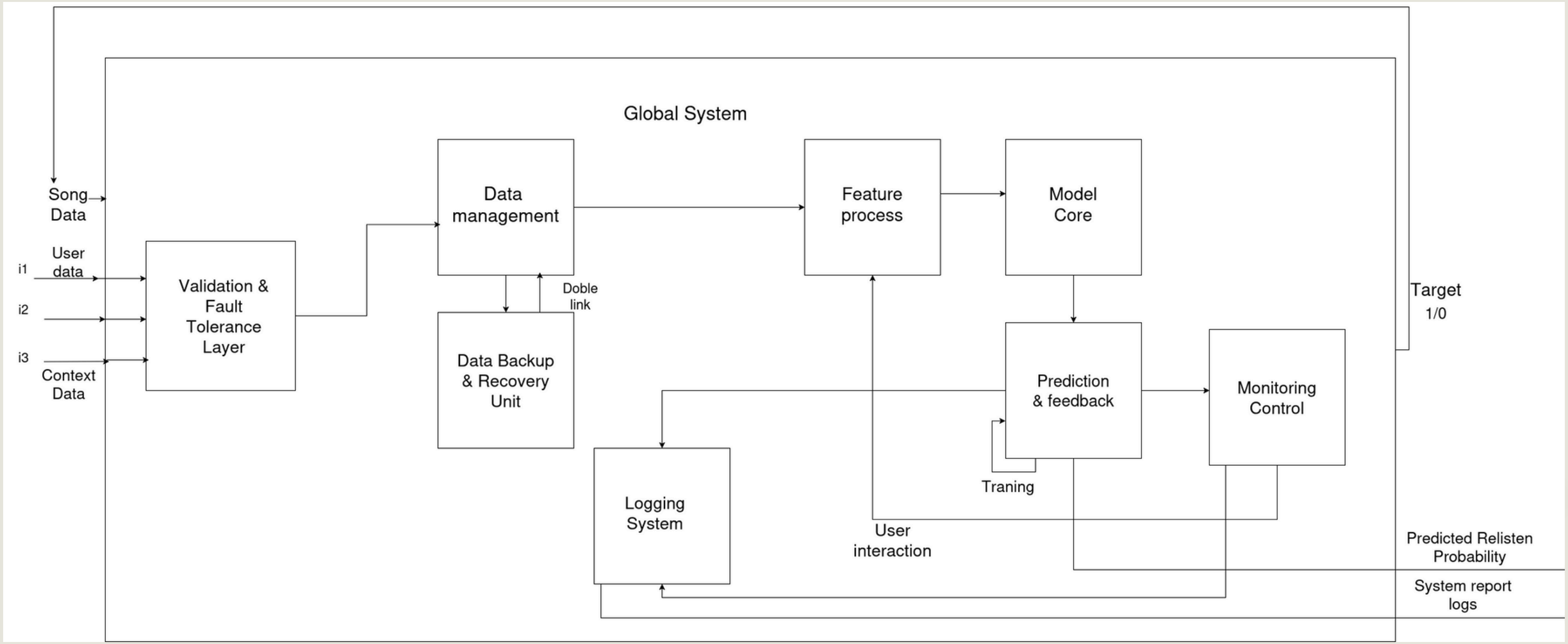
Songs: genre, artist, language, duration.

Time: when the interaction occurs.

Context: how the interaction happens (playlist, search, library).

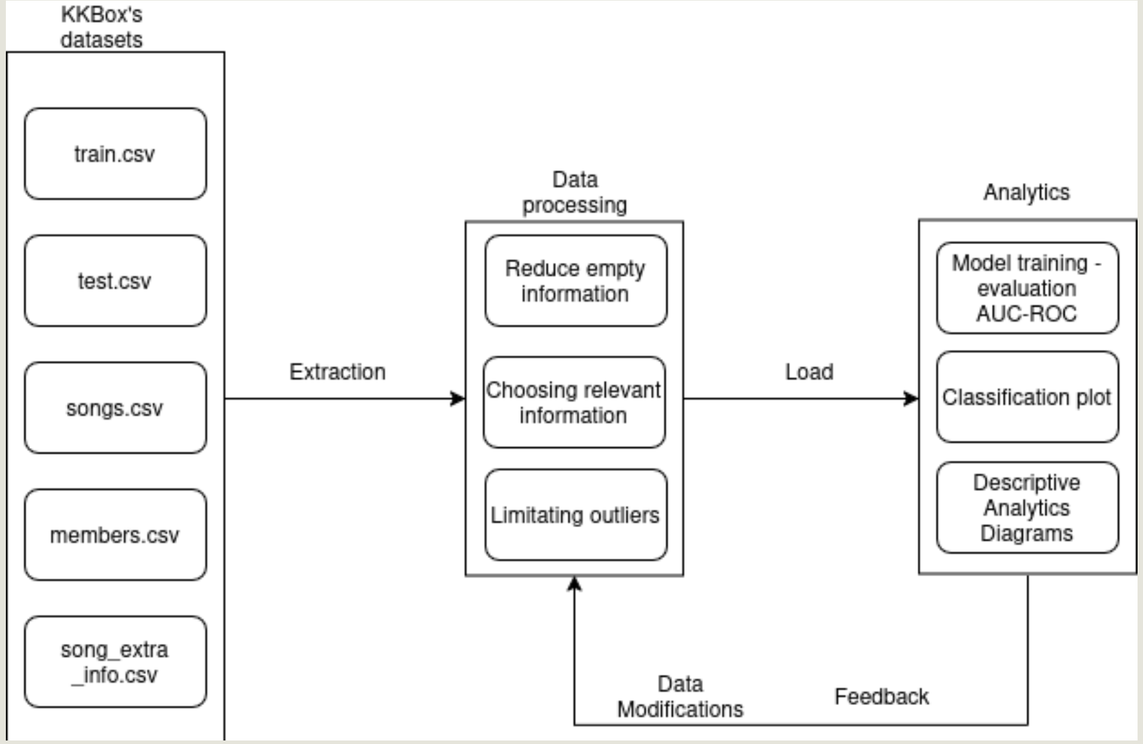


SYSTEM DESIGN:



PIPELINE:

The data is extracted from KKBox’s datasets.

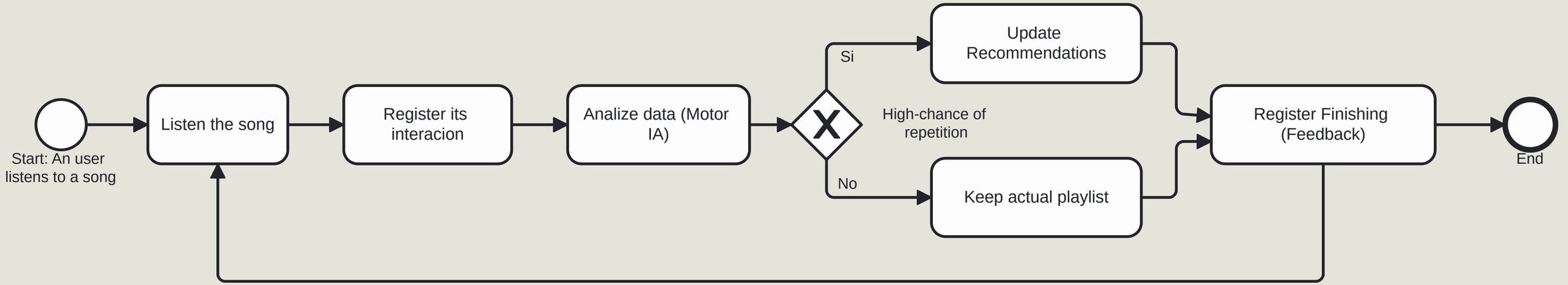


It must be processed limitating the outliers it presents and cleaning noise, like empty information, it the may contain.

The data is analyzed using charts to identify relations between columns and then is used to train the model.

According to the competition, the model must be evaluated using the AUC-ROC metric (Area Under the Curve - Receiver Operation Characteristic). Identifying the relation among TP and FP.

BUSINESS PROCESS MODEL AND NOTATION:



The BPMN diagram shows the flow through which a music application predicts whether a user will listen to a song again. The process begins when the user plays a track, and the application records this interaction. Then, the data is sent to the artificial intelligence engine, which analyzes the behavior and estimates the probability of repetition.

After the analysis, the system evaluates whether the calculated probability exceeds a defined threshold. If the value is high, the application updates the user's recommendations; if not, it keeps the current playlist. Finally, the system records the process outcome as feedback to improve the model.

System Robustness Principles

The system architecture follows robust design principles to ensure reliability, resilience, and adaptability across all components. A modular and layered structure isolates functions and prevents faults from propagating, while automated recovery routines handle errors without disrupting global operations.

Continuous monitoring, logging, and validation reinforce data integrity and enable proactive detection of instability. Together with secure data management and well-defined interfaces, these principles create a system capable of maintaining consistent performance even under uncertainty or variable input conditions.

- Modular, layered architecture
- Fault tolerance via recovery routines
- Monitoring, logging, validation layers
- Backup and data versioning
- Ensures system reliability despite data or model instability.

Quality Guidelines

Six Sigma: reduces noise, ensures stable processes.

ISO 27001: guarantees security, integrity, confidentiality of data.
Applied to dataset processing, model outputs, and system maintenance.

Risk Analysis



The system faces several risks. Corrupted, incomplete, or outdated datasets can lead to inaccurate predictions or even system failures, while shifting musical trends may reduce the model's relevance over time. To mitigate these issues, the pipeline incorporates validation checks, versioned backups, and continuous monitoring to detect inconsistencies early.

Main Risks:

- Corrupted or outdated data
- Performance loss due to evolving user behavior
- System instability due to noise or missing validation

Mitigation:

- Data checks, versioning, backups
- Retraining with recent data
- Monitoring system performance
- Progressive maintenance operations

METHODOLOGY: DATA.

Data Preparation:

- Merged KKBox datasets (members, songs, extra info, train).
- Cleaned invalid ages, imputed missing categories as “unknown”.
- Converted dates to datetime; removed duplicates; applied sparsity filter.
- Final dataset: 22 features, 7.37M rows + 50k reduced slice for testing.

METHODOLOGY:
MODEL 1.

ML Simulation:

- Model: Random Forest classifier.
- Selected 8 key predictors; label encoding for categorical features.
- Stratified 80/20 train–test split with fixed seed.
- Metrics: ROC-AUC (main), Accuracy (secondary).

METHODOLOGY:
MODEL 2.

Cellular Automata Simulation:

- 50×50 grid, binary states, Moore neighborhood.
- 300 iterations; 1–8 random events per iteration.
- Event types: play, activation, neighbor influence, noise.
- Activation threshold: ≥3 active neighbors.

RESULTS:

ML Scenario:

- Full dataset: ROC-AUC = 0.6634, Acc ≈ 0.63
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- Reduced slice: ROC-AUC = 0.7345, Acc = 0.7435

- Model performance improved with cleaner, smaller data.

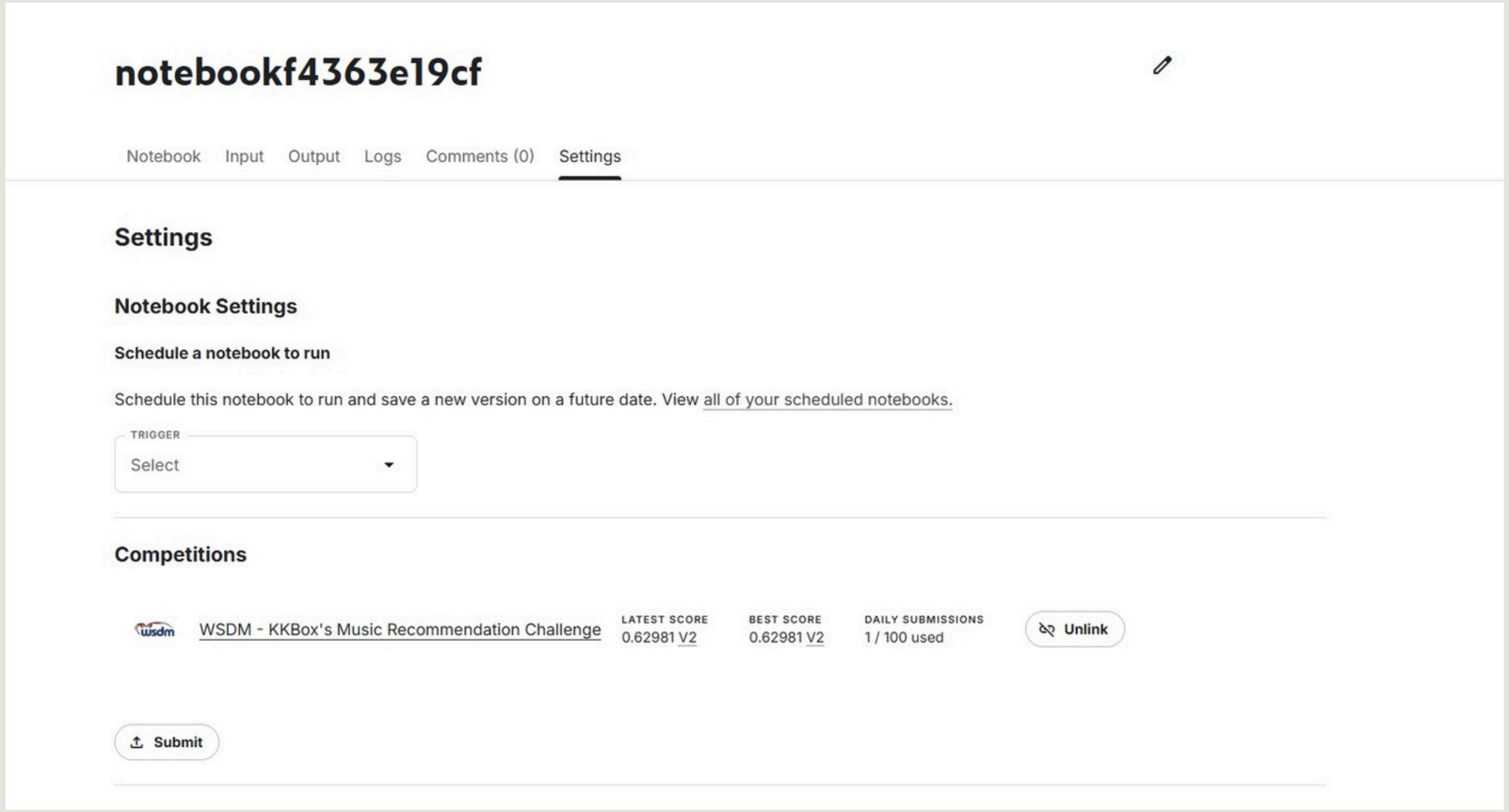
CA Scenario:

- Highly sensitive to small probability changes.
- Activation patterns inverted easily with minor noise shifts

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- Demonstrated chaotic/phase-like behavior near thresholds.

KAGGLE SCORE:



WE UPLOADED OUR MODEL TO KAGGLE PLATFORM, AND OBTAIN A 0.63 ON PRECISION PUNCTUACTION.

DISCUSSION:

ML Model

The ML model demonstrated a strong dependency on data quality and preprocessing.
Performance improved significantly after cleaning inconsistent categories and reducing noise, showing that stable feature distributions are essential for reliable replay prediction.
Although the Random Forest model produced acceptable results, its limitations suggest that more advanced algorithms and richer feature engineering are needed to capture the full complexity of user behavior.

CELLULAR AUTOMATA

In contrast, the Cellular Automata simulation revealed highly sensitive and chaotic dynamics. Small adjustments in event probabilities led to drastically different activation patterns, highlighting the fragile and unpredictable nature of local interaction-based propagation.
This behavior suggests that replay tendencies can emerge from nonlinear, unstable processes that traditional ML models may not fully capture.

CONCLUSIONS:

- Data quality is critical: Clean preprocessing significantly improves ML performance.
- Random Forest works, but performance varies with sparsity and categorical noise.
- CA model shows extreme sensitivity, useful for exploring propagation dynamics but unstable without calibration.
- Both approaches complement each other:
  - **ML**: Shows a realistic prediction
  - **CA**: Enables system-behavior exploration
- Future improvement: better categorical grouping, advanced models (XGBoost), CA decay and multi-state transitions.